Synthetic-Only 6D Inlet Pose Estimation and Robotic Docking for Megawatt EV Charging

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***Abstract*—Megawatt conductive charging for heavy-duty electric vehicles demands robotic docking under tight mechanical tolerances, yet acquiring real training data with accurate 6D ground truth is labor-intensive and must be repeated for each inlet revision or camera reconfiguration. This paper demonstrates that a *synthetic-only* monocular RGB pipeline, trained entirely from CAD-generated data without any real-image collection, can provide sufficiently accurate 6D inlet pose estimation for closed-loop robotic docking on real hardware. Annotated training data are generated with BlenderProc using photorealistic rendering, domain randomization, and sensor-effect injection. A YOLOv11-Pose network predicts inlet keypoints, and an EPnP+RANSAC backend recovers the 6D pose for visual servoing. On a real-image test set of more than 4300 automatically labeled frames, the final model achieves $1.77\,\mathrm{mm}$ planar translation error and 4.5° scalar rotation error. Over 300 autonomous trials on a UR5e testbench, the system achieves 94% alignment success and 78% plug-in success with a mean docking time of $30.5\,\mathrm{s}$. The remaining gap to full plug-in success lies in contact-sensitive final insertion, not in perception accuracy.**

***Index Terms*—sim-to-real, synthetic data, 6D pose estimation, keypoints, EPnP, visual servoing, robotic docking, megawatt charging**

I. Introduction

The electrification of heavy-duty transport is accelerating deployment of Megawatt Charging System (MCS) infrastructure, which imposes sub-centimeter mechanical tolerances during connector engagement [1], [2]. Manual plug-in at these power levels is operationally hazardous, motivating fully automated robotic docking [3], [4]. Prior automated solutions have relied on structured-light sensors, depth cameras, or fiducial markers [3], [5], none of which offer the cost and flexibility of a purely monocular RGB pipeline trained without any physical data collection. Acquiring real training data with accurate 6D ground truth is labor-intensive and must be repeated for each inlet revision or environment change; synthetic-only training addresses this directly by generating annotated data from CAD without any manual labeling or physical setup.

This paper presents an end-to-end synthetic-only pipeline that generates training data from CAD, detects sparse inlet keypoints with YOLOv11-Pose [6], recovers 6D pose with EPnP+RANSAC [7], and deploys the estimate in a closed-loop alignment-and-insertion controller. We evaluate key synthetic-data design choices, quantify transfer on more than 4300 automatically labeled real images, and validate the system through 300 autonomous docking trials on a UR5e testbench.

A. Contributions

1. A *synthetic-only* monocular RGB pipeline achieves $1.77\,\mathrm{mm}$ planar translation error and 4.5° rotation error on real images, sufficient for closed-loop megawatt EV docking without any real-image fine-tuning.
2. An empirical study of synthetic-data design choices, distance specialization, keypoint layout, sensor-effect injection, model size, and dataset scaling quantifies each factor's contribution to sim-to-real transfer.
3. Over 300 fully autonomous docking trials achieve 94% alignment and 78% plug-in success, establishing that insertion-stage contact dynamics, not perception, are the dominant remaining failure mode.

II. Related Work

A. Robotic Charging and Plug Insertion

Automated conductive charging has been studied using structured light and force-vision coupling [3], [5]. The broader survey by Saputra et al. [4] identifies pose estimation and plug insertion robustness as the two principal open challenges; no prior work achieved fully autonomous docking from camera input alone. Most directly related, Rakhmatulin et al. [8] apply a deep keypoint detector to an EV plug-in task using auto-annotation on real images; the present work differs in targeting the megawatt regime with a fully synthetic training pipeline and closed-loop hardware validation on a 6-DoF manipulator.

B. Monocular 6D Pose Estimation

Keypoint-based methods predict 2D projections of known 3D points and recover pose via PnP [7]. DOPE [9] established that synthetic-only training can transfer to real imagery when photorealistic rendering and domain randomization are combined. Direct-regression approaches such as GDR-Net [10] and its successor GDRNPP [11] achieve strong BOP benchmark accuracy through improved detection and geometry-guided refinement. Foundation-model approaches such as FoundationPose [12] offer zero-shot generalization via render-and-compare strategies, but carry inference overhead incompatible with the $25\,\mathrm{Hz}$+ loop required for visual servoing on a Jetson Orin AGX. This work prioritizes a lightweight single-stage pipeline whose single-forward-pass detection meets the $40\,\mathrm{ms}$ latency budget.

C. Synthetic Data and Sim-to-Real Transfer

BlenderProc [13] provides a photorealistic rendering pipeline with BOP-format annotation export and is used here for dataset generation. Domain randomization [14] underpins sim-to-real transfer; a large-scale benchmarking study [15] confirmed that texture diversity and HDR lighting are critical

for generalization, motivating our use of HDR environment maps and PBR material randomization. ParaPose [16] further demonstrates that careful tuning of randomization distributions is the primary lever for synthetic-only transfer, motivating the controlled ablations in Sec. V.

III. Methods

We propose a fully synthetic, monocular RGB pipeline for robust robotic docking of a megawatt EV charging connector. Starting from a CAD model of the inlet, photorealistic training images are generated automatically using domain randomization and sensor-effect injection, requiring no physical data collection. A single-stage YOLOv11-Pose network is trained on this synthetic dataset to detect inlet-specific keypoints, whose 2D–3D correspondences are fed to an EPnP+RANSAC solver to recover the 6D inlet pose in real time. The estimated pose drives a closed-loop visual servo controller that aligns the robot end-effector before commanding the final insertion. The complete approach is designed for embedded deployment and is validated end-to-end on a real UR5e test-bench. The following subsections detail each component of this pipeline.

A. System Overview

Fig. 1 summarizes the pipeline. Synthetic images from CAD train a YOLOv11-Pose network [6] to detect predefined inlet keypoints. The resulting 2D–3D correspondences are passed to an EPnP+RANSAC backend [7] to recover the 6D inlet pose, which drives the closed-loop alignment and insertion controller.

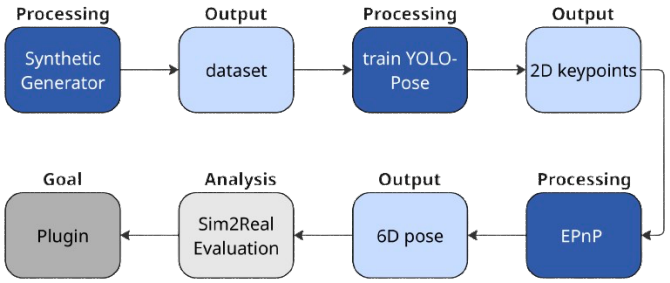


Fig. 1: Overview of the proposed docking system: CAD-based synthetic data generation, YOLOv11-Pose keypoint detection, EPnP+RANSAC pose estimation, and closed-loop visual servoing.

B. Problem Definition

Let $\mathbf{T}_{CI} \in SE(3)$ denote the inlet pose in the camera frame, with translation $\mathbf{t}_{CI}$ and rotation $\mathbf{R}_{CI} \in SO(3)$. The perception module estimates $\hat{\mathbf{T}}_{CI}$ from a single RGB image by detecting 2D keypoints $\{\hat{\mathbf{u}}_k\}_{k=1}^{K}$ for known 3D inlet points $\{\mathbf{p}_k\}_{k=1}^{K}$:

$$\hat{\mathbf{T}}_{CI} = \arg\min_{\mathbf{T}\in SE(3)} \sum_{k\in\mathcal{I}} \left\| \pi(\mathbf{T}\mathbf{p}_k) - \hat{\mathbf{u}}_k \right\|_2, \qquad (1)$$

where $\pi(\cdot)$ is the camera projection and $\mathcal{I}$ is the RANSAC inlier set. The estimate is transformed to the robot base frame via hand–eye calibration and used for Cartesian visual servoing toward the desired insertion pose $\mathbf{T}^{*}_{BI}$.



Fig. 2: UR5e testbench. The RGB camera is mounted on the end-effector and observes the vehicle-side inlet during alignment and insertion.

C. Pose Convention and Error Definitions

All poses are rigid transforms in $SE(3)$. The camera frame is right-handed with z forward, x right, and y down. Component-wise translation errors are

$$e_x = |\hat{x} - x^*|, \quad e_y = |\hat{y} - y^*|, \quad e_z = |\hat{z} - z^*|, \qquad (2)$$

and the planar translation error relevant to docking alignment is

$$e_{xy} = \sqrt{(\hat{x} - x^*)^2 + (\hat{y} - y^*)^2}. \qquad (3)$$

Rotational errors are reported as component-wise ZYX Euler-angle differences $(e_{\text{roll}}, e_{\text{pitch}}, e_{\text{yaw}})$ from $\mathbf{R}_{\text{err}} = \hat{\mathbf{R}}_{CI}\mathbf{R}_{CI}^{*\top}$, and as the scalar geodesic error

$$e_R = \cos^{-1}\left(\frac{\text{tr}(\mathbf{R}_{\text{err}}) - 1}{2}\right). \qquad (4)$$

D. Synthetic Data Pipeline

Training images are generated with BlenderProc [13] from a CAD model of the inlet. Each scene randomizes illumination (80 HDR environment maps), inlet surface material (60 PBR textures), background (100 indoor/outdoor images), and occluder placement (0–8 objects; occluders present in 60% of frames). Sensor-effect injection applies Gaussian blur ($\sigma \in [0, 0.08]$ px), exposure shifts (± 0.5 EV), JPEG compression (quality $\in [30, 95]$), chromatic aberration, and color jitter (brightness, contrast, and saturation $\in [0.9, 1.1]$) to narrow the sim-to-real gap. Camera viewing angles are sampled within $\pm 30^\circ$ of the inlet normal.

Two camera distance regimes are studied: short-range (0.5 m to 1.0 m), targeting the final alignment phase, and long-range (1.0 m to 2.0 m). Their impact is quantified in Sec. V.

E. Keypoint-Based Pose Estimation

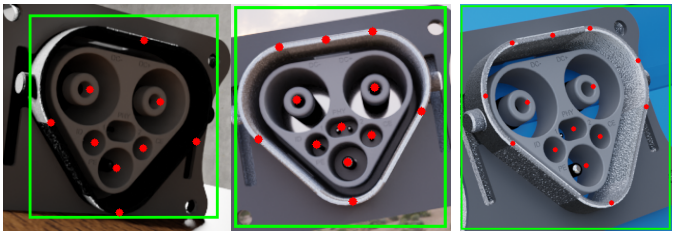
YOLOv11-Pose [6] jointly detects the inlet bounding box and regresses 2D keypoint locations in a single forward pass. The predicted keypoints are passed to EPnP [7], which provides an $O(n)$ closed-form solution well-suited to real-time control. RANSAC (100 iterations, 5 px reprojection threshold) provides outlier rejection; these parameters were chosen to



Fig. 3: Representative synthetic training images. Randomization spans viewpoint, distance, illumination, background, surface appearance, partial occlusion, and simulated sensor artifacts.

balance detection-noise robustness against the 40 ms perception budget. Keypoints below a confidence threshold of 0.5 are discarded before pose solving.

Keypoint count trades geometric redundancy against detection reliability. We evaluate layouts from 6 to 16 keypoints and select 14 keypoints for the deployed system (Sec. V).



(a) 9 keypoints (b) 12 keypoints (c) 14 keypoints

Fig. 4: Three of the evaluated keypoint layouts on the inlet CAD model. Keypoints are placed at geometrically distinctive features to maximize EPnP conditioning.

F. Implementation Details

YOLOv11-X was trained for 50 epochs with AdamW (lr $= 1 \times 10^{-3}$, cosine decay, batch 8) using standard mosaic, flip, HSV, and scale jitter augmentation. All hyperparameters are fixed across ablations unless stated otherwise. Key parameters of the deployed system are listed in Table I.

TABLE I: Key parameters of the deployed perception and control pipeline.

Parameter	Value
Image resolution	640 × 640 px
Keypoint confidence threshold	0.5
RANSAC threshold / iterations	5 px / 100
Optimizer / LR / schedule	AdamW / 10^{-3} / cosine
Training epochs / batch	50 / 8
Camera calibration error	≈0.3 px
Control loop frequency	10 Hz
Alignment thresholds	5 mm, 3°
Stability interval	1 s
Perception latency	40 ms

G. Closed-Loop Visual Servoing and Insertion

Docking proceeds in two stages. In the *alignment stage*, the latest valid pose estimate drives a Cartesian visual servo loop that reduces translation and rotation error toward a pre-insertion target. Validity is continuously monitored via keypoint confidence and EPnP+RANSAC reprojection checks. Once the error remains within the alignment thresholds for a 1 s stabilization interval, the controller enters the *insertion stage*, advancing along the connector axis under constrained motion with fixed orientation. The trial aborts on a validity failure, threshold violation, or stage timeout. This two-stage design explicitly separates perception-limited failures from contact-stage failures.

IV. Experimental Setup

A. Evaluation Metrics

Pose accuracy is reported via (e_x, e_y, e_z), $(e_{\text{roll}}, e_{\text{pitch}}, e_{\text{yaw}})$, e_{xy}, and e_R as defined in Sec. III-C. For task-level pose consistency we use ADD-S [17]:

$$\text{ADD-S} = \frac{1}{|\mathcal{M}|} \sum_{\mathbf{p} \in \mathcal{M}} \min_{\mathbf{q} \in \mathcal{M}} \left\| (\hat{\mathbf{R}}\mathbf{p} + \hat{\mathbf{t}}) - (\mathbf{R}\mathbf{q} + \mathbf{t}) \right\|_2, \quad (5)$$

where $\mathcal{M}$ is the set of 3D model points. We report both the standard ADD-S (all axes) and a task-adapted variant that excludes z. The exclusion is motivated by two task-specific properties: monocular depth along the insertion axis is inherently ambiguous in the short-range regime, producing errors that dominate the metric; and depth accuracy at alignment time is not required, since insertion completion is detected through electrical contact. Both variants use the standard 10%-diameter threshold ($d = 112$ mm, threshold = 11.2 mm).

For robotic trials we report alignment success, plug-in success, and mean docking time, with 95% Wilson score confidence intervals.

B. Ground-Truth Pose and Auto-Labeling

Ground-truth inlet pose is obtained from a rigid panel carrying 12 ArUco markers with a known board-to-inlet transform $\mathbf{T}_{BI}$:

$$\mathbf{T}_{CI} = \mathbf{T}_{CB}\, \mathbf{T}_{BI}, \quad (6)$$

where $\mathbf{T}_{CB}$ is estimated from detected marker corners. Frames with fewer than four detected markers or a board reprojection error above 1.0 px are discarded; the median retained reprojection error is 0.35 px. The panel is used exclusively for ground-truth recovery and auto-labeling; it plays no role in the deployed pipeline.

The evaluation set contains more than 4300 frames collected under varying illumination, backgrounds, and distractor configurations with a Basler ace 2 camera (a2A1920-51gcBAS, 6 mm lens). Real images used for the fine-tuning study (Sec. V-G) are disjoint from this evaluation set.

C. Occlusion Test Protocol

Occlusion is defined as the fraction of the inlet projection covered by a physical object placed between the camera and the inlet. Six controlled levels are tested: 0%, 10%, 20%, 30%, 40%, and 50%. Fifty trials are run per level with randomized object placement; e_{xy} and plug-in success rate are reported.

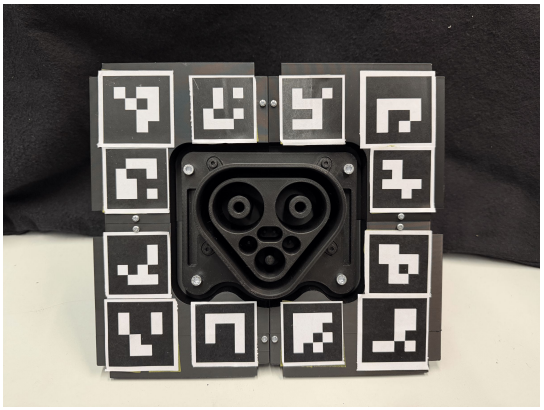


Fig. 5: Auto-labeling setup. The ArUco panel provides ground-truth inlet pose for evaluation and keypoint label generation.

Autonomous docking is enabled only when at least 50% of the inlet is visible; detection capability at higher occlusion levels was characterized separately in non-autonomous trials.

D. Hardware and Runtime

Synthetic data were generated on an Apple M4 Pro (24 GB unified memory). Deployment and training ran on an NVIDIA Jetson Orin AGX (64 GB RAM). The robot is a UR5e with an end-effector-mounted Basler ace 2 camera. YOLOv11-X was deployed via TensorRT at $\approx$25 FPS, with 40 ms end-to-end perception latency. The onboarding time from CAD to a deployable model is approximately 30 min for keypoint definition, 35 h for 10k-image dataset generation, and 37 h for training on the embedded platform; cloud training is expected to reduce this total by a factor of 5–10.

V. Experimental Results

This section presents the experimental evaluation of the proposed pipeline in two stages. We first characterize the perception system in isolation through a series of controlled vision ablations, examining the impact of camera distance specialization, dataset scale, keypoint count, sensor-effect injection, and model size on synthetic-to-real transfer accuracy (Secs. V-A through E). We then evaluate the complete docking system on real hardware, reporting sim-to-real transfer metrics, fine-tuning costs, closed-loop robot trial outcomes, failure mode analysis, occlusion robustness, and generalization to a second connector standard (Secs. V-F through J).

A. Distance Specialization

Short-range training (0.5 m to 1.0 m) improves near-field accuracy at the cost of long-range performance (Table II). This ablation uses 1k images and YOLOv11-L to isolate the effect of the distance range; the full 10k / YOLOv11-X configuration is used for deployment.

B. Dataset Scaling and Final Configuration

Table III reports pose accuracy as the dataset size grows from 3k to 10k images (14 keypoints, YOLOv11-X, short-range). Accuracy improves consistently, with the largest gains

TABLE II: Distance specialization (1k images, YOLOv11-L). Short-range sampling improves near-field accuracy critical to the alignment stage.

Range	Simulation			Real			
	e_{xy} (mm)	e_z (mm)	e_R (°)	e_{xy} (mm)	e_z (mm)	e_R (°)	ADD-S (%)
Short	3.1	23.7	9.7	3.6	30.1	9.4	91.2
Long	3.6	20.5	7.1	4.1	28.7	9.6	89.4
Full	3.3	22.4	10.6	4.0	29.3	9.6	89.6

occurring between 3k and 7.5k images and diminishing returns thereafter. The 10k configuration is selected as the final setup.

TABLE III: Effect of dataset size (14 keypoints, YOLOv11-X, short-range). Diminishing returns appear beyond 7.5k images.

Size	Simulation			Real			
	e_{xy} (mm)	e_z (mm)	e_R (°)	e_{xy} (mm)	e_z (mm)	e_R (°)	ADD-S (%)
3k	2.5	14.7	7.6	2.5	15.9	7.6	93.2
5k	2.3	13.9	5.4	2.3	14.4	7.4	93.9
7.5k	2.3	13.5	5.1	2.1	13.9	6.9	95.2
10k	1.63	13.0	3.88	1.77	13.5	4.5	96.8

C. Keypoint Count

Fig. 6 shows that accuracy improves up to 14 keypoints and then saturates, indicating that sufficient geometric constraints for stable EPnP recovery are reached at that count. The 14-keypoint layout is used in all subsequent experiments.

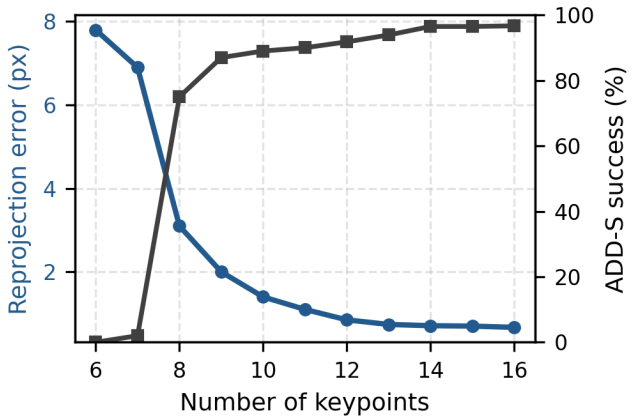


Fig. 6: Real-image reprojection error and ADD-S vs. keypoint count (10k images, YOLOv11-X). Performance saturates beyond 14 keypoints.

D. Sensor-Effect Injection

Table IV shows that sensor-effect injection substantially reduces real-image error compared to noise-free rendering. The improvement is most pronounced on real images, confirming that injection targets the sim-to-real domain gap rather than synthetic test noise. Both configurations use 1k images and YOLOv11-L; injection is the sole varied factor.

TABLE IV: Impact of sensor-effect injection (1k images, YOLOv11-L, short-range). Real-image accuracy improves substantially with injection enabled.

Config	Simulation			Real			
	e_{xy} (mm)	e_z (mm)	e_R (°)	e_{xy} (mm)	e_z (mm)	e_R (°)	ADD-S (%)
No injection	4.5	28.2	12.3	7.0	42.1	12.1	78.7
Injection	3.1	23.7	9.7	3.6	30.1	9.4	91.2

E. Model Size and Runtime

Table V compares YOLOv11-L and YOLOv11-X on the final 10k / 14-keypoint / short-range configuration. YOLOv11-X reduces all error metrics at the cost of $\approx 2\times$ lower throughput; both models operate above the 10 Hz control frequency on the Jetson Orin AGX. YOLOv11-L achieves 50 Hz and YOLOv11-X achieves 25 Hz, both satisfying the real-time requirement for UR5e control.

TABLE V: Model-size trade-off (10k images, 14 keypoints, short-range). Both variants satisfy the real-time constraint.

Model	Simulation			Real			
	e_{xy} (mm)	e_z (mm)	e_R (°)	e_{xy} (mm)	e_z (mm)	e_R (°)	ADD-S (%)
YOLOv11-L	2.7	16.7	7.6	3.2	18.1	8.2	93.4
YOLOv11-X	1.6	13.0	3.9	1.7	13.5	4.5	96.8

F. Sim-to-Real Transfer

Table VI compares simulation and real-image errors for the final deployed model. The retention ratio ($E_{\text{sim}}/E_{\text{real}}$) is a descriptive summary; values near 100% indicate small transfer degradation. Planar translation retains 92.1% accuracy. Pitch exhibits the largest rotational degradation (85.3%) due to limited out-of-plane keypoint separation in the inlet geometry ($\approx$10 mm), which reduces EPnP conditioning for that axis. Standard ADD-S results (including z) are provided in Table VII for benchmark comparability; the ADD-S cumulative curve is shown in Fig. 7.

TABLE VI: Sim-to-real comparison for the final deployed model. Retention near 1.0 indicates small transfer degradation.

Metric	Sim	Real	Retention
e_{xy} (mm)	1.63	1.77	92.1 %
e_R (°)	3.88	4.50	86.2 %
x (mm)	1.20	1.35	88.9 %
y (mm)	1.00	1.18	84.7 %
z (mm)	13.0	13.5	96.3 %
Roll (°)	0.70	0.80	87.5 %
Pitch (°)	2.90	3.40	85.3 %
Yaw (°)	2.50	2.80	89.3 %

G. Effect of Real-Data Fine-Tuning

The primary objective of this work is to demonstrate that a *synthetic-only* pipeline is sufficient for deployment in settings where real training data cannot be collected, for instance, during the initial commissioning of a new inlet design, or when a camera is remounted without vehicle access. To characterize the performance ceiling achievable with limited real data, and to contextualize where the synthetic-only model sits relative to that ceiling, we additionally evaluate fine-tuning using small sets of auto-labeled real images (Sec. IV-B).

TABLE VII: Standard ADD-S (all axes) vs. task-adapted ADD-S (excluding z) for the final model.

Split	ADD-S standard (%)	ADD-S task-adapted (%)
Simulation	85.1	97.2
Real	79.3	96.8

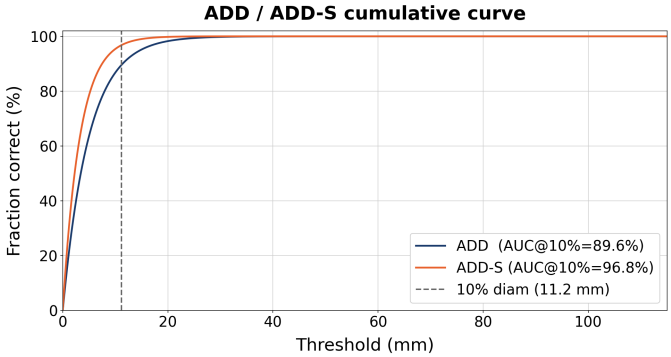


Fig. 7: ADD-S (task-adapted) cumulative accuracy on the real-image test set. The dashed line marks the 10%-diameter threshold (11.2 mm).

Five training configurations are evaluated on the same disjoint real-image test set: (i) synthetic-only; synthetic pre-training fine-tuned with (ii) 150, (iii) 300, and (iv) 500 auto-labeled real images; and (v) real-only training on 10k images. Results are shown in Fig. 8.

Fine-tuning consistently improves ADD-S over the synthetic-only baseline, confirming that real images carry complementary information not captured by the synthetic distribution. Gains are measurable at 150 images and saturate near 300–500 images, at which point the model approaches the real-only performance ceiling. As an additional reference point, DOPE [9] trained on the same synthetic dataset achieves 38% ADD-S@10%, shown as a dashed baseline in Fig. 8. The substantially lower score is consistent with DOPE's cuboid-corner keypoint formulation: when the inlet is observed frontally, the rear cuboid corners project in close proximity to their front-face counterparts, producing near-degenerate PnP correspondences on a largely texture-less planar surface. Importantly, the synthetic-only model already occupies the large majority of the achievable performance range: the gap between synthetic-only and the saturated fine-tuned model is small relative to the gap between noise-free synthetic training and the synthetic-only baseline (compare Table IV and Fig. 8). This confirms that the engineering quality of the synthetic pipeline, not the absence of real data, is the primary determinant of transfer performance.

All robotic docking experiments reported in Sec. V-H use the synthetic-only model exclusively, corresponding to the

target deployment scenario.

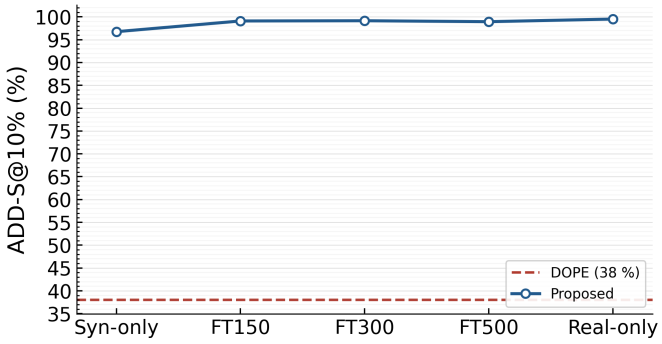


Fig. 8: ADD-S (task-adapted) on the real-image test set for synthetic-only, fine-tuned (150–500 real images), and real-only models. The dashed red line shows DOPE [9] trained on the same data as a reference; its lower score reflects the unsuitability of cuboid-corner keypoints for a frontally observed planar inlet. The synthetic-only model occupies the large majority of the achievable range; fine-tuning offers measurable but bounded further improvement.

H. Closed-Loop Robot Validation

Each trial starts from a randomized end-effector pose: lateral offsets $\in [-25, 25]$ cm in x and y, initial distance $\in [0.4, 1.0]$ m, and orientation offsets $\in [-15°, 15°]$ for roll, pitch, and yaw. The inlet pose is fixed; the robot pose is re-randomized between runs. All 300 trials execute fully autonomously. Trials were distributed over three consecutive days (100 per day) and multiple times of day to sample natural lighting variation; performance was consistent across sessions.

The system achieves **94% alignment success** (Wilson 95% CI: 90.6–96.2%) and **78% plug-in success** (Wilson 95% CI: 73.0–82.3%) with a mean successful docking time of 30.5 s. Among plug-in failures, insertion/contact failure accounts for 51% and perception-related failures (dropout and pose rejection) account for 49% combined. These figures are summarized in Fig. 9.

Fig. 10 shows the closed-loop convergence of e_{xy} over successful runs. Two stages are visible: a coarse camera-based alignment that rapidly reduces large lateral offsets, followed by a connector-level refinement that converges within the insertion tolerance.

I. Failure Mode Analysis

Failures are classified as *perception dropout* (detection loss), *pose rejection* (EPnP+RANSAC validity failure), or *insertion/contact failure* (alignment achieved but physical plug engagement failed). As Fig. 9 shows, insertion/contact sensitivity is the dominant remaining limitation, confirming that the perception pipeline is not the primary bottleneck in the tested configuration.

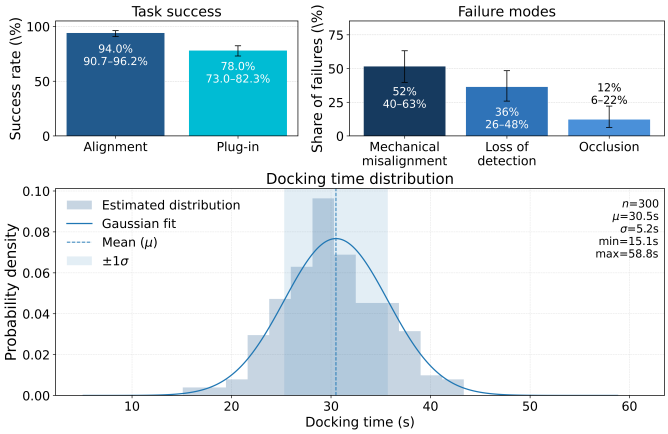


Fig. 9: Summary of 300 autonomous docking trials. Alignment succeeds in 94% of trials; plug-in succeeds in 78%. Insertion/contact failure is the largest single plug-in failure category. Loss-of-detection failures are partially mitigable through last-known-pose fallback strategies, which would allow alignment to continue from the most recent valid estimate rather than aborting the trial.

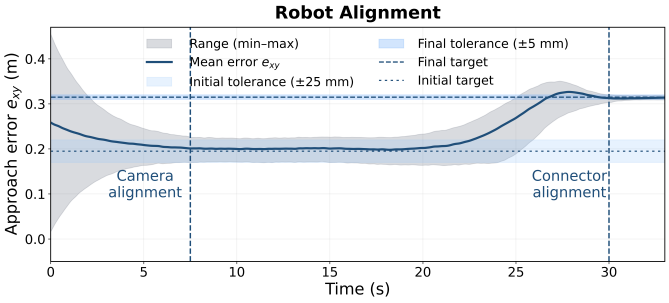


Fig. 10: Convergence of e_{xy} during successful docking trials. The mean trajectory and percentile envelope are shown. Both the coarse alignment and connector-level refinement stages are visible.

J. Occlusion Robustness

Fig. 11 reports e_{xy} and plug-in success rate at six occlusion levels (0–50%, 50 trials each). Performance degrades gracefully up to 30% occlusion; beyond this, the reduction in the number and spatial spread of visible keypoints degrades EPnP conditioning even when the inlet remains detectable.

K. Generalization to a Second Connector Standard

To assess whether the synthetic-only pipeline generalizes beyond the MCS inlet, the full on-boarding procedure was applied to a CCS2 connector without any architectural modification. Starting from a new CAD model, keypoint definition, BlenderProc scene configuration, and training were completed within 3 days of engineering effort. Table VIII reports task-adapted ADD-S on a real-image evaluation set collected with the same Basler ace 2 camera setup. The result is comparable to the MCS configuration, confirming that the pipeline is not tuned to a specific inlet geometry and that the on-boarding time estimate of Sec. IV-D is achievable in practice. No closed-loop

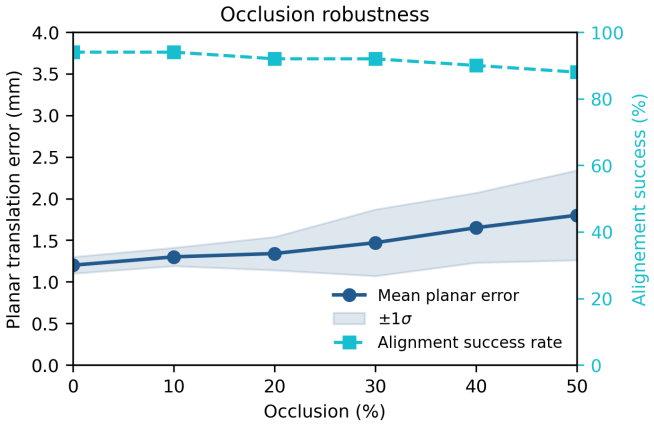


Fig. 11: Occlusion robustness over 50 trials per level. e_{xy} (left axis) and plug-in success (right axis) vs. occlusion fraction. Autonomous docking is gated to $\leq$50% occlusion for pose observability.

robot trials were conducted on CCS2; perception-only transfer is reported here.

TABLE VIII: Task-adapted ADD-S on real images for both connector standards. CCS2 used an identical pipeline with no architectural changes.

Connector	ADD-S task-adapted (%)
MCS	96.8
CCS2	94.0

VI. Discussion

The results establish that synthetic-only training provides accurate near-field pose estimation for robotic docking when the synthetic distribution is matched to the deployment regime. Short-range camera sampling and sensor-effect injection are the two most impactful design choices, collectively reducing real-image e_{xy} from $7.0\,\mathrm{mm}$ (noise-free, full-range) to $1.77\,\mathrm{mm}$ in the final configuration. This improvement confirms that the quality of synthetic data engineering, not the volume of real data, is the primary lever for sim-to-real transfer in this task.

Plug-in success requires that residual misalignment at the moment of insertion falls within the mechanical capture range of the MCS connector. The MCS specification defines a circular alignment tolerance of $2\,\mathrm{mm}$ radial and $2°$ angular for reliable self-guided engagement without active compliance [1], [2]. Let the residual planar error at the end of the alignment stage be ϵ_{xy} and the residual angular error be ϵ_R. A trial proceeds to successful plug-in if and only if both conditions hold simultaneously:

$$\epsilon_{xy} \leq \delta_t \quad \text{and} \quad \epsilon_R \leq \delta_\theta, \tag{7}$$

where $\delta_t = 2\,\mathrm{mm}$ and $\delta_\theta = 2°$ are the connector's hard mechanical tolerances. The perception pipeline achieves a mean e_{xy} of $1.77\,\mathrm{mm}$ and e_R of $4.5°$; while the planar translation is within tolerance on average, the rotational error distribution has a non-negligible tail above δ_θ, directly explaining why a fraction of trials that achieve the $5\,\mathrm{mm}$ / $3°$ alignment gate still fail at insertion. The 16-percentage-point gap between alignment (94%) and plug-in (78%) success is therefore consistent with the rotational error distribution rather than a perception failure.

Two hardware interventions are expected to close this gap substantially. First, a passive compliant mechanism at the end-effector introduces a lateral accommodation range of $3\,\mathrm{mm}$ to $5\,\mathrm{mm}$ and a rotational float of $2°$ to $4°$ [5], effectively widening δ_t and δ_θ in (7) without requiring any change to the perception or control software. Second, switching the final insertion motion from position control to current-limited torque control allows the robot to comply with contact forces during engagement, tolerating residual angular misalignment through reactive yielding. Either intervention alone is expected to raise plug-in success above 90%; their combination, which is the standard approach in industrial connector-insertion systems, should approach the alignment success ceiling of 94%.

From a deployment standpoint, the synthetic-only approach eliminates the need for manual real-image annotation when adapting to a new inlet geometry or camera configuration. The complete onboarding path from CAD to a deployable model requires no physical data collection and takes approximately $72\,\mathrm{h}$ on the embedded Jetson Orin AGX platform; cloud training is expected to reduce this total by a factor of 5–10.

Pitch degrades more than yaw in sim-to-real transfer because the inlet keypoints span only $\approx 10\,\mathrm{mm}$ in the out-of-plane direction, reducing EPnP conditioning for that axis. This asymmetry could be partially mitigated through keypoint placement on deeper geometric features or geometry-aware training loss weighting.

The occlusion analysis confirms that the 50% visibility gate is appropriate: below this threshold, keypoint spatial distribution degrades sufficiently to compromise EPnP conditioning regardless of whether the inlet itself remains detectable.

VII. Conclusion

This paper demonstrates that a synthetic-only pipeline, combining CAD-based data generation, domain randomization, sensor-effect injection, single-stage keypoint detection, and EPnP+RANSAC pose recovery, is sufficient for real-time closed-loop megawatt EV charging docking. Over 300 autonomous trials on a UR5e testbench, the system achieves 94% alignment success and 78% plug-in success with no real training data. The dominant remaining failure mode is insertion-stage contact dynamics, identifying compliance-aware insertion control as the most impactful next step toward fully autonomous megawatt charging.

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